CODE ALPHA - DATA ANALYTICS INTERNSHIP

Task 2: Exploratory Data Analysis (EDA)

Internship ID: CA/DF1/323302

Name: Laiba Abbas

Domain: Data Analytics

Batch: 20 Oct 2026 - 20 Nov 2026

Key Findings:

1. No duplicate values found
2. Max Loss: -6599.98 at 70% discount
3. Top 20 loss orders: discount 0.4 to 0.8
4. Pattern: High Discount = High Loss

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CODEALPHA INTERNSHIP - TASK 2 REPORT

Task 2: Exploratory Data Analysis (EDA) - Superstore Dataset

Name: Laiba Abbas
Internship ID: Awaiting (Not Received Yet)
Dataset: Sample - Superstore (9994 Rows, 21 Columns)
Tool Used: Microsoft Excel
Date: October 2026

1. OBJECTIVE:
To perform EDA on Superstore dataset to check data quality and find business insights.

2. STEPS PERFORMED:

A) Duplicate Check:
- Method: Selected full data -> Data Tab -> Remove Duplicates
- Result: Microsoft Excel shows "No duplicate values found"
- Conclusion: 0 Duplicates. Dataset is clean.
- Proof: Screenshot 1 (Page 2)

B) Missing Values Check:
- Method: Ctrl+G -> Go To Special -> Blanks
- Result: No blank cells found
- Conclusion: 0 Missing Values

C) Profit & Loss Analysis (Main Finding of Task 2):
- Method: Sorted Profit column from Smallest to Largest
- Result Found:
  * Maximum Loss: -6599.98 at 70% Discount (0.7)
  * Second Loss: -3839.99 at 50% Discount (0.5)
  * Third Loss: -3701.89 at 80% Discount (0.8)
- Pattern: All top 20 loss orders have discount 0.4 to 0.8 (40%-80%)
- Proof: Screenshot 2 (Page 3)

3. KEY INSIGHT FROM EDA:
- High Discount is the main reason for loss
- When discount > 0.2 (20%), profit becomes negative
- Company is giving up to 80% discount which is causing huge loss

4. RECOMMENDATION:
- Limit discount to maximum 20% for all products
- Specially reduce discount on Furniture category

5. CONCLUSION:
Task 2 EDA completed successfully. Data has no duplicates or missing values.
Main business problem identified is wrong discount strategy.

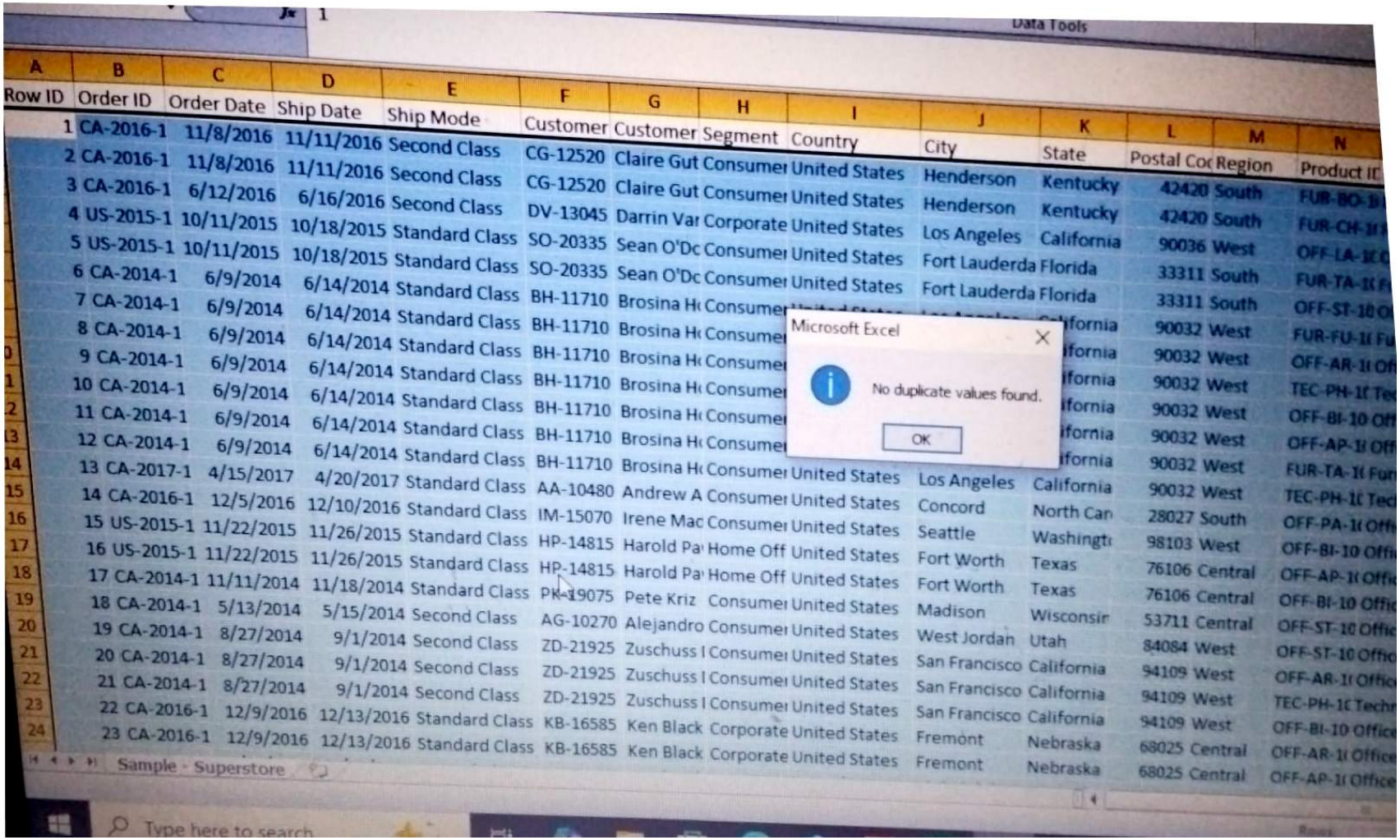
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TASK 2 - SCREENSHOT PROOF 1

A) Duplicate Check - No Duplicate Values Found

This is the proof for Task 2 EDA - Duplicate Check Step



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